Unit 12, Lesson 3: Solving Problems Involving Surface Area of Right Circular Cylinders

Benchmarks

MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

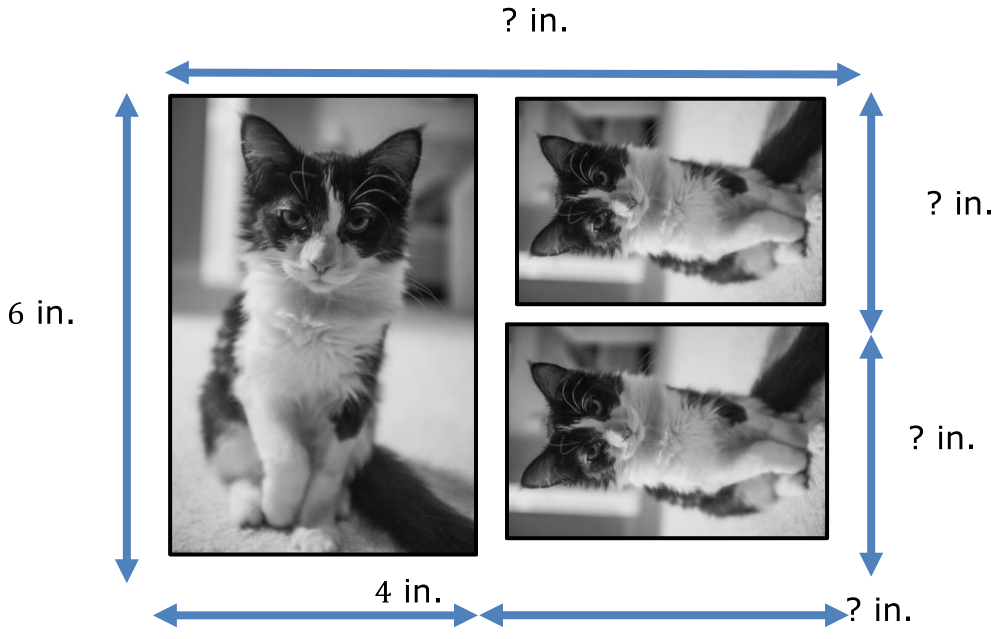
MA.7.GR.2.2 Solve real-world problems involving surface area of right circular cylinders.

Learning Targets

- ☐ I can draw the net of a right circular cylinder.
- ☐ I can solve problems involving the surface area of a right circular cylinder.

12.3.1 Warm-Up: Photographs

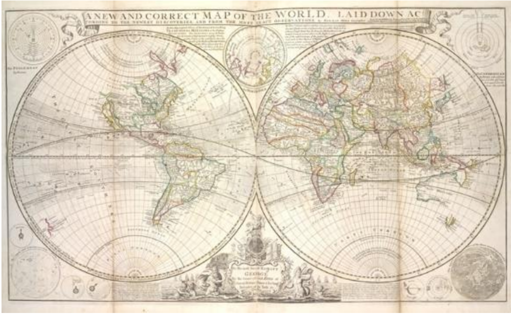
1. A photographer wants to print a photograph and two smaller copies on the same rectangular sheet of paper. The smaller picture is proportional to the larger picture. Dimensions are given in inches (in.). The diagram is not drawn to scale.



What are the dimensions of the small photographs?

12.3.2 Guided Instruction: Finding the Surface Area of Right Circular Cylinders

1. For her final art project, Courtney uses an old-world map to decorate a right-cylindrical pedestal. She wraps the map around the lateral area of the pedestal.



   a. Draw a net of the pedestal and label the part of the net where the map will be located.

   b. The length of the map is 16 feet (ft), and its height is 6.5 ft. The radius of the pedestal is 2.55 ft. Find the surface area of the pedestal. Use 3.14 for π and round your answer to the nearest hundredth.

2. With your partner, write a formula that could be used to find the surface area of a right circular cylinder.

3. Ask two classmates to share and explain the formula they wrote. Record their formula in the table. *You are the only person who should write in the first column of the table.* Have the person verify that you correctly recorded their formula by writing their initials.

Formula	Initials

4. Review the formulas from your classmates with your partner. Determine if there is more than 1 formula that can be used to find the surface area of a right circular cylinder. If necessary, revise the formula you and your partner wrote.

Surface Area of a Cylinder

$SA = 2B + Ph$, where B is the area of the base, P is the perimeter of the base and h is the height

Or

$SA = 2\pi rh + 2\pi r^2$, where r is the radius of the circular base and h is the height

5. The Push Pop candy, a creative spin on the lollipop, was very popular in the 1990s. It is cylindrical in shape, with a diameter of 2 centimeters (cm) and a height of 8 cm.

Find the surface area of the Push Pop candy. Use $\frac{22}{7}$ for π and round to the nearest hundredth.

12.3.3 Guided Practice: Solving Problems Involving Surface Area of Right Circular Cylinders

1. Cheese is often made into a cylindrical shape. Some cheeses are encased in paraffin wax after being formed into a cylindrical shape. The wax specially for coating cheeses aides in preventing unwanted mold growth and retains moisture. An example of what the wax coating looks like when removed from around the cheese is shown in the picture.

The circumference of a type of Gouda cheese shown is 9.5 in. and it has a height of 5 in.

How much wax coating will be needed to encase the cheese? Use $\frac{22}{7}$ for π and round your answer to the nearest hundredth.

2. A checker piece is in the shape of a right circular cylinder and has a diameter of 1.75 in. Each checker piece that is sold in a vintage game set is individually wrapped snug in 6.9 in^2 of material. What is the height, in inches, of each individual checker piece? Use 3.14 for π and round your answer to the nearest hundredth.

3. Tim plans to give his wife a roll of toilet paper as a gag gift for their first wedding anniversary since paper is the traditional 1 year wedding anniversary gift. If the roll of toilet paper is 10 cm tall by 12 cm wide, how much wrapping paper, in cm^2 will he need to cover the gift? Use $\frac{22}{7}$ for π and round your answer to the nearest hundredth.

4. Amoré gave her homeroom teacher a cylindrical jar with a hot chocolate mix inside. The jar she used was 4.2 in. wide by 6 in. tall. The gift was such a hit that she wants to give her 5 other teachers the same hot chocolate mix and tightly wrap each jar in plastic to preserve the ingredients. How much plastic will she need in total to wrap the 5 jars? Leave your answer in terms of π and include units.

12.3.4 Wrap-Up: Ticket Out the Door

Consider the formula, $SA = 2\pi rh + 2\pi r^2$, to complete the following.

1. A hydration lotion is sold in a cylindrical container. The lotion ships in a box, but for quality control purposes the jar is wrapped in vacuum sealed plastic before being placed into a shipping box. The dimensions of the jar are 7.8 in. tall by 3.39 in. wide. How much plastic material is used to wrap around 60 individual jars of lotion? Use 3.14 for π and round your answer to the nearest whole number.